

Quotient-Robust ENN Bayesian Optimization

Learning Which Decision-Space Directions the Objective Can Distinguish

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Abstract

We consider Bayesian optimization of expensive black-box objectives over high-dimensional continuous, quantized, and mixed-variable domains. Our starting point is a local structural hypothesis: objective variation is dominated by a low-dimensional block pseudometric whose null classes form approximately invariant fibres. We introduce quotient-robust epistemic nearest-neighbour upper confidence bound (QR-ENN-UCB), which maintains a confidence set of plausible pseudometrics and optimizes an upper confidence bound jointly over candidate points and geometries. Each round evaluates one objective value; coordinate perturbations are used only to construct the finite acquisition domain. We establish an additive simple-regret cost for approximate fibres, an impossibility result for certifying null directions from finitely many unrestricted observations, a finite-candidate regret decomposition into prediction uncertainty, geometry uncertainty, candidate coverage, and quotient misspecification, and a conditional nonparametric rate governed by quotient dimension rather than ambient dimension. We then give a systems design for quantized model parameters and map the resulting method onto implicit candidate descriptors, bit-packed storage, and accelerator-resident ENN scoring. The framework separates the statistical claim of dimension reduction from the representation and computational mechanisms used to realize it.

1 Introduction

Bayesian optimization (BO) allocates a small budget of expensive function evaluations by combining a surrogate model with an acquisition rule [1, 2]. Its application to high-dimensional parameter spaces is limited by two distinct costs. Statistically, nonparametric prediction and candidate coverage deteriorate with dimension. Computationally, conventional surrogates and acquisition optimizers can become more expensive than the black-box interface can tolerate. Existing high-dimensional BO methods address these costs through local trust regions, random or adaptive subspaces, additive structure, and specialized sampling schemes [3, 4, 6, 5]. These methods still require the optimizer to specify, assume, or learn which directions are relevant.

We study the case in which relevance is local and may be degenerate. Within a trust region, the objective may vary substantially along a small number of directions while remaining nearly constant on large sets of configurations. A positive-definite metric can rescale directions but cannot express exact invariance. We instead use a pseudometric: zero-distance classes represent configurations that the objective cannot distinguish up to a controlled error. The induced quotient is allowed to vary between trust regions.

Learning this geometry from the evaluations used for BO creates a model-selection problem. A single regularized metric estimate can assign zero weight to an untested direction and subsequently prevent the acquisition function from testing it. QR-ENN-UCB retains a set of geometries compatible with the observations. Its acquisition function is optimistic over this set, and its regret bound contains an explicit geometry-disagreement term. Dimension reduction is therefore claimed only when the geometry set contracts and the fibre approximation error remains controlled.

Contributions. The paper makes four methodological contributions.

1. It formulates local dimension reduction as approximate factorization through an objective-dominating pseudometric and isolates the resulting fibre-misspecification cost.
2. It separates a domination set, which preserves valid objective bounds, from an identification set, which supports model selection and approximate elimination of directions. A finite-sample impossibility theorem shows why these roles cannot be conflated.
3. It introduces QR-ENN-UCB and proves a finite-candidate regret decomposition whose terms expose the statistical requirements for ambient-dimension-independent behavior.
4. It gives an implementation design for quantized high-dimensional models that distinguishes the logical optimization domain, coordinate proposal kernel, ENN metric, and physical encoding.

The present results do not constitute a global no-regret theorem. Such a theorem additionally requires contraction of geometry uncertainty, adaptive candidate-coverage guarantees, and control of trust-region transitions. We state these missing conditions explicitly rather than absorbing them into an oracle assumption.

1.1 Problem formulation

Let

$$\mathcal{X} = \prod_{\ell=1}^L \mathcal{X}_{\ell}, \quad D = \sum_{\ell=1}^L n_{\ell},$$

be a compact continuous, discrete, quantized, or mixed decision space. At round t , an algorithm chooses $x_t \in \mathcal{X}$ based on the filtration $\mathcal{F}_{t-1}$ and receives

$$Y_t = F(x_t) + \varepsilon_t, \quad \mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}, x_t) = 0, \quad (1)$$

where the noise is conditionally σ -sub-Gaussian. We maximize F ; minimization reverses signs. At each round the algorithm constructs a finite candidate set $A_t \subset \mathcal{X}$, evaluates an acquisition function on A_t , and queries one selected point. The candidate set may depend on the past, subject to the measurability conditions used in the confidence analysis.

1.2 Quantized domains and proposal kernels

For quantized numerical parameters, the logical search space is a Cartesian product of finite code alphabets,

$$\mathcal{C} = \prod_{\ell=1}^L \mathcal{Z}_{\ell}^{n_{\ell}}, \quad \mathcal{Z}_{\ell} = \{0, \dots, 2^{b_{\ell}} - 1\}, \quad q_{\ell} : \mathcal{Z}_{\ell}^{n_{\ell}} \rightarrow \mathbb{R}^{n_{\ell}}, \quad (2)$$

where q_{ℓ} decodes a block using its quantization format and scale. The numerical geometry is defined on decoded values. Domains combining continuous, integer, ordinal, or categorical variables fall under mixed-variable optimization [7, 8].

For $c, z \in \mathcal{C}$, define the generalized Hamming distance

$$d_H(c, z) = \sum_{\ell=1}^L \sum_{r=1}^{n_{\ell}} \mathbf{1}\{c_{\ell r} \neq z_{\ell r}\}. \quad (3)$$

The support-restricted proposal kernel samples s coordinate indices and changes only those coordinates, so its support lies in the Hamming ball $\mathcal{N}_s(c) = \{z : d_H(c, z) \leq s\}$. We refer to this as fixed-cardinality coordinate mutation or a support-restricted coordinate proposal. A second proposal kernel may perturb coordinates across the full vector. Hamming trust regions and local acquisition search are standard in categorical and combinatorial BO [8, 9]; the decoded metric introduced below additionally records the numerical magnitude and objective relevance of code changes.

Both proposal kernels define finite acquisition domains; neither constructs a finite-difference gradient. QR-ENN-UCB predicts and acquires over the resulting candidates and evaluates one point per round. Section 13 distinguishes this feedback model from zeroth-order gradient estimation.

2 Local quotient model

For decoded block values $q_\ell(x) \in \mathbb{R}^{n_\ell}$ define

$$\begin{aligned} D_\ell(x, z) &= \|q_\ell(x) - q_\ell(z)\|_2^2, \\ G(g) &= \text{diag}(g_1 I_{n_1}, \dots, g_L I_{n_L}), \\ m_g(x, z) &= \sum_{\ell=1}^L g_\ell D_\ell(x, z), & d_g(x, z) &= m_g(x, z)^{1/2}. \end{aligned}$$

for $g \in \mathbb{R}_+^L$. Zero coordinates are intentional: d_g is a pseudometric. Its equivalence classes define a quotient map

$$\pi_g : U \longrightarrow B_g = U / \sim_g, \quad x \sim_g z \iff d_g(x, z) = 0,$$

inside a local region $U \subset \mathcal{X}$. A block weight is therefore not merely a length scale. At zero it asserts that all allowed movement in that block is observationally invisible up to a declared error.

Definition 1 (admissible quotient geometry). For $\alpha \in (0, 1]$, $L_F > 0$, and $\eta \geq 0$, a geometry g is (α, L_F, η) -admissible for F on U if

$$|F(x) - F(z)| \leq L_F d_g(x, z)^\alpha + 2\eta \quad (x, z \in U). \quad (4)$$

If $F(x) = f(\pi_g(x)) + e(x)$, f is α -Hölder under d_g , and $\|e\|_{\infty, U} \leq \eta$, then (4) follows. Conversely, when $d_g(x, z) = 0$, (4) bounds every fibre's objective diameter by 2η . Define the exact local half-oscillation

$$\eta_U(g) = \frac{1}{2} \sup_{b \in B_g} \left\{ \sup_{x \in \pi_g^{-1}(b)} F(x) - \inf_{x \in \pi_g^{-1}(b)} F(x) \right\}. \quad (5)$$

The quotient terminology has its standard mathematical meaning: a function constant on equivalence classes factors uniquely through the corresponding quotient. Here the equivalence classes are the zero-distance fibres of d_g , and the residual in (5) quantifies approximate rather than exact factorization.

Proposition 2 (cost of quotienting). *Let $f_g(b)$ be the midpoint of the range of F on the fibre over b . For any queried lifts $x_t \in \pi_g^{-1}(b_t)$,*

$$\max_{x \in U} F(x) - \max_{t \leq T} F(x_t) \leq \max_{b \in B_g} f_g(b) - \max_{t \leq T} f_g(b_t) + 2\eta_U(g). \quad (6)$$

The constant 2 is sharp when the lifts are uncontrolled.

Proof. By construction, $|F(x) - f_g(\pi_g(x))| \leq \eta_U(g)$. Apply opposite sides of this inequality to an optimizer and the best queried lift. Opposite fibre endpoints attain equality. $\square$

Proposition 2 isolates the approximation cost of dimension reduction: an approximate quotient exchanges ambient dimension for a simple-regret floor. Consequently, $\eta_U(g)$ must appear in any guarantee based on an inexact quotient.

2.1 Proposal geometry versus quotient geometry

Sampling on an ellipsoid or hypersphere changes the proposal measure and may improve radial concentration or candidate coverage; cylindrical sampling uses this effect in high-dimensional BO [5]. Such a proposal does not by itself reduce the number of directions along which F varies. QR-ENN-UCB samples metric balls or spheres under each plausible geometry, while dimensional reduction arises only from the null classes of the pseudometric.

3 Identifiability of quotient geometry

A natural baseline chooses the minimum-complexity metric that majorizes observed response differences. For local pairs E_t , one might solve

$$\begin{aligned} (\hat{g}_t, \hat{\xi}_t) \in \arg \min_{g, \xi \geq 0} \quad & c^\top g + C \sum_{(i,j) \in E_t} \omega_{ij} \xi_{ij}, \\ \text{subject to} \quad & h_{ij} \leq \langle g, D_{ij} \rangle + \xi_{ij}, \end{aligned} \tag{7}$$

where $D_{ij} = (D_1(x_i, x_j), \dots, D_L(x_i, x_j))$ and h_{ij} is a noise-corrected squared response separation. The ℓ_1 objective encourages zero blocks and is computationally attractive. It is not, by itself, a confidence procedure. If two blocks are correlated in the sampled configurations, the linear program can assign all variation to either one. If a block has not been tested, the cost term sets it to zero precisely because there is no evidence.

The failure is statistical, not a tuning defect.

Theorem 3 (finite data cannot certify a null direction). *Let $U \subset \mathbb{R}^D$ be nonempty and open, let $S = \{x_1, \dots, x_n\} \subset U$ be finite, and let ℓ be a coordinate block. There exist smooth objectives $F_0, F_1 : U \rightarrow \mathbb{R}$ such that*

- (i) $F_0(x_i) = F_1(x_i)$ for every $x_i \in S$;
- (ii) block ℓ is null for F_0 ;
- (iii) block ℓ is non-null for F_1 and the two objectives have different maxima.

Consequently, under the same observation-noise law, no rule based only on observations at S can distinguish the two objectives or certify that block ℓ is globally null.

Proof. Choose a point z and a small ball $V \subset U$ disjoint from S . Let φ be a smooth bump supported in V whose value changes in block ℓ . Take $F_0 = 0$ and $F_1 = \varphi$. They agree on S , while only F_1 varies in block ℓ and has a positive maximum. Since their response distributions on S coincide, every observation-measurable decision has the same law under both. $\square$

Remark 4 (requirements for exact elimination). An exact zero requires additional structure: known invariance, exhaustive coverage at the relevant resolution, an equivalence test with a declared smoothness class and power, or an explicit prior-risk assumption. In the absence of such structure, a strictly positive reference geometry preserves reachability and small weights represent approximate local irrelevance rather than exact invariance.

4 Confidence sets for quotient geometry

We use a set-valued geometry estimate and separate evidence that guarantees objective domination from evidence that identifies a parsimonious geometry.

4.1 Domination set

Suppose a simultaneous event provides bounds

$$|Y_i - F(x_i)| \leq b_i \quad (i < t). \quad (8)$$

For a proposed (α, L_F, η) , define

$$a_{ij} = \left(\frac{(|Y_i - Y_j| - b_i - b_j - 2\eta)_+}{L_F} \right)^{2/\alpha}. \quad (9)$$

Let $\Gamma = [0, \bar{g}_1] \times \cdots \times [0, \bar{g}_L]$ encode physically meaningful maximum scales and put

$$\mathcal{G}_t^{\text{dom}} = \{g \in \Gamma : a_{ij} \leq \langle g, D_{ij} \rangle \text{ for all } (i, j) \in E_t\}. \quad (10)$$

This is a polytope. Scale is retained in g for statistical domination; candidate motion can use a separately normalized copy.

Proposition 5 (coverage of the true geometry). *On event (8), if $g_\star \in \Gamma$ is (α, L_F, η) -admissible on U , then $g_\star \in \mathcal{G}_t^{\text{dom}}$ for every t whose pair set lies in U .*

Proof. The reverse triangle inequality and (8) give $(|Y_i - Y_j| - b_i - b_j - 2\eta)_+ \leq (|F(x_i) - F(x_j)| - 2\eta)_+$. Apply (4), raise both sides to $2/\alpha$, and use $d_{g_\star}^2 = \langle g_\star, D_{ij} \rangle$. $\square$

The proposition survives adaptive BO locations because it only needs a simultaneous noise event; time-uniform sub-Gaussian confidence sequences or a summable allocation of failure probabilities can provide such an event. But $\mathcal{G}_t^{\text{dom}}$ has a crucial limitation: its constraints are lower bounds on weighted distances. Increasing a coordinate of g preserves feasibility. Domination data can prove that a block is needed, but cannot prove that a block is unnecessary.

4.2 Identification set

Let $\mathcal{I}_t \subseteq \Gamma$ be a confidence set from a predeclared identification model, and require

$$\mathbb{P}(g_\star \in \mathcal{I}_t \text{ for every } t) \geq 1 - \delta_{\text{geom}}. \quad (11)$$

The active geometry set is

$$\mathcal{G}_t = \mathcal{G}_t^{\text{dom}} \cap \mathcal{I}_t. \quad (12)$$

If this intersection is empty, at least one of L_F , η , the noise envelope, or the identification model must be relaxed. Projection to a metric with unsupported zero block weights would invalidate the coverage event.

We consider three constructions for $\mathcal{I}_t$.

1. **Finite model family.** Predeclare geometries $g^{(1)}, \dots, g^{(J)}$, run residual tests, and eliminate a model only after a time-uniform test rejects its advertised domination and calibration guarantees. This construction is the most direct and is closest to bandit model selection [16].
2. **Local variogram model.** Use randomized, predictable block perturbations to estimate the block-wise second moment of objective increments. Confidence intervals for those moments induce a box or ellipsoid for g . This can identify small directions, but its conclusion is local and requires curvature control.
3. **Structural invariance.** A decoder or simulator may declare transformations known to leave the evaluated system's behaviour unchanged. These directions can be quotiented exactly without reward evidence.

The point estimate (7) remains useful as one representative of $\mathcal{G}_t$, not as the whole state. Additional representatives include coordinate extremizers

$$g_{t,\ell}^{\min} \in \arg \min_{g \in \mathcal{G}_t} g_\ell, \quad g_{t,\ell}^{\max} \in \arg \max_{g \in \mathcal{G}_t} g_\ell,$$

and geometries returned by acquisition maximization. Their disagreement is the uncertainty that prevents self-confirming collapse.

4.3 Local identification through block variograms

The following calculation explains why blockwise perturbations are informative without turning the optimizer into a finite-difference method.

Lemma 6 (block variogram). *Suppose F is C^2 near c with $\|\nabla^2 F\|_{\text{op}} \leq H$. Let $\Delta = (\Delta_1, \dots, \Delta_L)$ have independent centred blocks with $\|\Delta_\ell\| = r_\ell$ and $\mathbb{E}\Delta_\ell\Delta_\ell^\top = (r_\ell^2/n_\ell)I$. As $r = (\sum_\ell r_\ell^2)^{1/2} \rightarrow 0$,*

$$\mathbb{E}\{F(c + \Delta) - F(c)\}^2 = \sum_{\ell=1}^L \frac{\|\nabla_\ell F(c)\|_2^2}{n_\ell} r_\ell^2 + O(\|\nabla F(c)\| H r^3 + H^2 r^4). \quad (13)$$

Proof. Taylor-expand at c . Centring and block independence remove cross terms, and isotropy gives the displayed quadratic term. The first-order term is $O(\|\nabla F(c)\| r)$ and the Hessian remainder is $O(H r^2)$, so their cross term and the squared remainder give the stated error. $\square$

Thus a natural infinitesimal target is

$$g_\ell^\circ(c) = \frac{\|\nabla_\ell F(c)\|_2^2}{n_\ell}. \quad (14)$$

Support-restricted coordinate proposals can estimate (13) over many rounds by comparing a new reward to a stored centre or local anchor. The selected candidate evaluation still contributes to optimization; there is no mandatory antithetic mate. With noisy objectives, a reused noisy anchor induces dependence and a squared-increment noise bias. A rigorous estimator must use a replicate-averaged anchor, independent anchor observations, or a martingale analysis that includes that dependence.

Small gradient at one point is not a fibre certificate. If $\|\nabla^2 F\|_{\text{op}} \leq H$ on a radius- ρ region and a confidence bound gives $\|\nabla_\ell F(c)\| \leq u_\ell$, then

$$\sup_{x \in U} \|\nabla_\ell F(x)\| \leq u_\ell + H\rho. \quad (15)$$

If Z is a proposed set of null blocks and every two points on one proposed fibre can be joined by a path whose total motion in block ℓ is at most R_ℓ , then

$$2\eta_U(g) \leq \sum_{\ell \in Z} R_\ell(u_\ell + H\rho). \quad (16)$$

Equation (16) provides an approximate-collapse certificate: it combines statistical uncertainty, curvature, and trust-region scale. It also explains why a block may become negligible only after the region contracts.

5 ENN surrogate and calibration

For a query x and geometry g , retrieve the k nearest prediction observations under d_g . For noise estimates s_i^2 , define inverse-distance weights

$$a_{i,g}(x) = \{\epsilon + s_i^2 + \lambda d_g(x, x_i)^2\}^{-1}, \quad w_{i,g}(x) = \frac{a_{i,g}(x)}{\sum_j a_{j,g}(x)}, \quad (17)$$

and

$$\widehat{F}_{t,g}(x) = \sum_i w_{i,g}(x) Y_i, \quad N_{\text{eff}}(t, g, x) = \frac{1}{\sum_i w_{i,g}(x)^2}. \quad (18)$$

Write $r_{t,g,k}(x) = \max_{i:w_{i,g}(x)>0} d_g(x, x_i)$.

ENNX’s internal epistemic and aleatoric scales are useful acquisition features. They become a Bayesian posterior only under a generative model, and they become frequentist confidence bounds only after a coverage argument. We use “ENN-UCB” to name the decision rule, not to assert that the existing ENNX scale already has the following guarantee.

Theorem 7 (cross-fitted ENN interval). *Fix g , a query x , and neighbour weights that are independent of the prediction-fold response noises. Suppose g is (α, L_F, η_g) -admissible and the selected noises are independent σ -sub-Gaussian variables. Then, with probability at least $1 - \delta$,*

$$|\widehat{F}_{t,g}(x) - F(x)| \leq C_{t,g}(x; \delta), \quad (19)$$

where

$$C_{t,g}(x; \delta) = L_F r_{t,g,k}(x)^\alpha + 2\eta_g + \sigma \sqrt{2 \log(2/\delta) \sum_i w_{i,g}(x)^2}. \quad (20)$$

Proof. For every selected neighbour, admissibility bounds the deterministic difference from $F(x)$ by $L_F r_{t,g,k}(x)^\alpha + 2\eta_g$. Convex averaging preserves this bound. The weighted noise sum is sub-Gaussian with proxy variance $\sigma^2 \sum_i w_{i,g}(x)^2$. $\square$

Strict independence can be obtained with a geometry/training stream and a prediction stream whose locations are committed before their responses are observed. For Theorem 7, g , the candidate pool, and the centre used to generate that pool must be measurable without the prediction-stream responses. Retrospectively splitting a single adaptive BO trajectory is insufficient: later locations encode earlier noise. Cross-fitting can rotate genuinely valid streams, while an all-data predictor needs a separate predictable-weight or martingale proof. This independence condition is essential because reusing outcomes both to choose g and to report an interval under g can cause systematic undercoverage.

Conditionally on a sigma-field under which a finite candidate pool A_t , a finite representative geometry set Q_t , and all prediction weights are fixed independently of prediction-fold noise, a union bound uses $\delta/(|A_t||Q_t|)$. Finiteness alone is not enough when the set itself depends on that noise. A continuous geometry set requires either a uniform empirical-process bound or a finite net plus a continuity error. Our finite implementation uses a small Q_t and reports the resulting discretization error; continuous-set optimization is not treated as exact.

6 Quotient-robust acquisition and regret

Let $Q_t \subset \mathcal{G}_t$ be a finite representative set. For each $g \in Q_t$, define

$$u_{t,g}(x) = \widehat{F}_{t,g}(x) + C_{t,g}(x), \quad \ell_{t,g}(x) = \widehat{F}_{t,g}(x) - C_{t,g}(x). \quad (21)$$

The quotient-robust upper envelope and computable geometry disagreement are

$$U_t(x) = \max_{g \in Q_t} u_{t,g}(x), \quad \Omega_t(x) = \max_{g \in Q_t} u_{t,g}(x) - \min_{g \in Q_t} u_{t,g}(x). \quad (22)$$

The acquisition rule is

$$x_t \in \arg \max_{x \in A_t} U_t(x). \quad (23)$$

Only x_t is evaluated. Maximizing over g implements optimism over structural uncertainty without imposing a posterior distribution on metrics.

Proposition 8 (one-step robust regret). *Let $x_U^* \in \arg \max_{x \in U} F(x)$ and $x_{A_t}^* \in \arg \max_{x \in A_t} F(x)$. Define the candidate-pool gap*

$$\Delta_t^{\text{pool}} = F(x_U^*) - F(x_{A_t}^*).$$

Suppose $g_ \in Q_t$ and Theorem 7's interval holds for all $x \in A_t$ under g_* . Then the selection (23) satisfies*

$$F(x_U^*) - F(x_t) \leq \Delta_t^{\text{pool}} + 2C_{t,g_*}(x_t) + \Gamma_t(x_t) \leq \Delta_t^{\text{pool}} + 2C_{t,g_*}(x_t) + \Omega_t(x_t), \quad (24)$$

where

$$\Gamma_t(x) = U_t(x) - u_{t,g_*}(x).$$

Proof. At the pool optimizer, $F(x_{A_t}^*) \leq u_{t,g_*}(x_{A_t}^*) \leq U_t(x_{A_t}^*)$. Selection gives $U_t(x_{A_t}^*) \leq U_t(x_t)$, while the true-geometry lower interval gives $F(x_t) \geq \ell_{t,g_*}(x_t)$. Hence

$$F(x_{A_t}^*) - F(x_t) \leq U_t(x_t) - \ell_{t,g_*}(x_t) = \Gamma_t(x_t) + 2C_{t,g_*}(x_t).$$

Add the pool gap. Since u_{t,g_*} is one member of the finite family, $\Gamma_t \leq \Omega_t$. $\square$

Proposition 8 gives the central decomposition. A point estimate suppresses Ω_t operationally without eliminating geometry uncertainty statistically. Robust acquisition retains this term and allocates evaluations where plausible geometries disagree. If the geometries make similar predictions at selected points, Ω_t is small even when their raw weights differ; prediction disagreement is therefore more relevant than parameter distance.

If $g_* \notin Q_t$ but Q_t is an ε_t -net of a confidence set containing it, and $u_{t,g}(x)$ is K_t -Lipschitz in g on that set, the same argument adds at most $2K_t\varepsilon_t$. This is the geometry-discretization term. Near zero weights, neighbour identities can change discontinuously, so K_t may be large. We therefore include coordinate extremes and observed neighbour-order breakpoints in the representative set.

Summing (24) gives the conditional cumulative-regret statement

$$R_T \leq \sum_{t=1}^T \Delta_t^{\text{pool}} + 2 \sum_{t=1}^T C_{t,g_*}(x_t) + \sum_{t=1}^T \Omega_t(x_t), \quad (25)$$

up to geometry discretization and trust-region transition terms. A sublinear theorem therefore requires all three sums to be controlled. Geometry coverage alone is not enough.

6.1 Intrinsic-dimension dependence

Let q denote a local covering dimension under the true pseudometric, not the count of nonzero ambient coordinates. Suppose that, conditionally on the selected region, the prediction history is quasi-uniform enough that

$$r_{t,g_*,k}(x_t) \leq C_r(k/t)^{1/q} \quad (26)$$

and the normalized weights satisfy $\sum_i w_{i,g_*}(x_t)^2 \leq C_w/k$. These are substantive sampling assumptions; adaptive BO does not grant them for free.

Corollary 9 (conditional quotient rate). *Under (26), exact fibres $\eta_{g_*} = 0$, bounded logarithmic factors, and $k_t \asymp t^{2\alpha/(2\alpha+q)}$, the true-geometry ENN radius obeys*

$$C_{t,g_*}(x_t) = \tilde{O}\left(t^{-\alpha/(2\alpha+q)}\right), \quad (27)$$

and its contribution to (25) is

$$\tilde{O}\left(T^{1-\alpha/(2\alpha+q)}\right). \quad (28)$$

Proof. The Hölder bias is $O((k/t)^{\alpha/q})$ and the stochastic term is $\tilde{O}(k^{-1/2})$. Balancing them gives the stated k_t and radius. Sum the resulting power law. $\square$

Ambient dimension D has disappeared from this conditional statistical term, but it can re-enter through candidate generation, geometry identification, Ω_t , or a poor block family. This is the precise sense in which quotient BO could scale, and also the list of ways it can fail.

7 Candidate-set construction and coverage

For a centre c , metric radius ρ , and coordinate-change budget s , let $d_{\text{supp}}(x, c)$ count the coordinates whose values differ. On the quantization-code domain $\mathcal{C}$, this is d_H from (3); in a continuous decoded chart it is $\|q(x) - q(c)\|_0$. Define the intersection of a pseudometric ball and a support ball,

$$\mathcal{T}_g(c, \rho, s) = \{x \in \mathcal{X} : d_g(x, c) \leq \rho, d_{\text{supp}}(x, c) \leq s\}. \quad (29)$$

The support ball controls how many variables change, while d_g controls the decoded magnitude and objective-relevant geometry of those changes. A continuous candidate can be formed by drawing a support-restricted direction z on a unit sphere or in a unit ball and applying a scaled lift $A_g z$ satisfying $A_g^\top G(g) A_g \approx I$. In the quantized backend, the analogue is bounded coordinate mutation inside $\mathcal{N}_s(c)$, followed by decoding for metric evaluation. For zero or tiny weights, step sizes must be capped by block radii and a reference metric; otherwise an inverse square root would produce unbounded coordinate changes.

The robust pool is a union

$$A_t = A_t^{\text{ref}} \cup \bigcup_{g \in Q_t} A_{t,g}, \quad A_{t,g} \subset \mathcal{T}_g(c_t, \rho_t, s_t). \quad (30)$$

Candidate descriptors can share seeds across geometries so that differences are attributable to scaling rather than Monte Carlo noise. A_t^{ref} uses a fixed strictly positive reference geometry and prevents an incorrect learned family from making part of the decision space unreachable.

The pool term has a simple dimension-dependent interpretation. Suppose the candidate law under $g_\star$ assigns mass at least $c\epsilon^{q/\alpha}$ to the ϵ -optimal set in the current region. For M independent true-geometry candidates,

$$\mathbb{P}(\Delta_t^{\text{pool}} > \epsilon) \leq \exp(-M c \epsilon^{q/\alpha}). \quad (31)$$

Thus, with probability $1 - \delta$,

$$\Delta_t^{\text{pool}} \leq \left(\frac{\log(1/\delta)}{cM} \right)^{\alpha/q}. \quad (32)$$

This elementary bound explains why large cheap candidate pools matter. It also explains why a sphere alone is insufficient: the exponent improves only when the proposal reaches a q -dimensional near-optimal set under a valid geometry.

8 QR-ENN-UCB algorithm

We call the proposed method **QR-ENN-UCB**: quotient-robust epistemic nearest-neighbour upper confidence bound. The analysis uses independent geometry/training and prediction streams as specified in Theorem 7. An all-data implementation is a practical approximation whose adaptive calibration is evaluated empirically in Section 11.

Algorithm 10 (QR-ENN-UCB). 1. **Localize.** Choose the active region U_t , centre c_t , metric radius ρ_t , and coordinate-change budget s_t from trust-region state.

2. **Update evidence.** Add valid secant constraints to $\mathcal{G}_t^{\text{dom}}$ and update the predeclared identification set $\mathcal{I}_t$. If their intersection is empty, relax the structural model or restart the local region.
3. **Represent geometry.** Build Q_t from the point-LP solution, the reference metric, coordinate extremizers, surviving finite models, and any geometry that maximized the previous acquisition. Cap $|Q_t|$ by merging metrics with indistinguishable predictions.
4. **Generate candidates.** Using shared seeds, sample bounded coordinate proposals inside each $\mathcal{T}_g(c_t, \rho_t, s_t)$ and add reference-geometry and restart candidates.
5. **Predict.** Run ENN neighbour retrieval and prediction for every $(x, g) \in A_t \times Q_t$. Form calibrated $u_{t,g}(x)$ and the robust envelope $U_t(x)$.
6. **Acquire.** Select $x_t \in \arg \max_{x \in A_t, g \in Q_t} u_{t,g}(x)$. Break ties by larger geometry disagreement, then by a deterministic seed order.
7. **Evaluate once.** Query Y_t , append the observation with its preassigned fold, and update success, coverage, geometry-disagreement, and region-transition diagnostics.

We record the acquisition-maximizing geometry alongside the selected configuration. If a candidate geometry repeatedly selects configurations whose rewards violate its intervals or whose empirical regret exceeds its candidate bound, it becomes eligible for elimination. Balancing or elimination methods for bandit model selection provide a template for this logic [16]; directly importing their regret theorem would still require each geometry-specific ENN learner to satisfy its advertised bound.

8.1 Trust-region updates

TuRBO supplies a practical success/failure mechanism for local BO regions [3]. We expand a region only when all of the following hold:

1. incumbent improvement is statistically credible;
2. held-out ENN coverage is not below target;
3. Ω_t is small near the proposed expanded boundary;
4. the fibre certificate (16) remains below its budget.

The region contracts when local coverage fails or curvature invalidates the geometry, and restarts when no member of $\mathcal{G}_t$ is admissible at a useful η . These tests distinguish objective failure from surrogate failure and geometry failure.

Different regions may have different quotients. On an overlap, two metrics g_a, g_b are compatible at distortion κ if

$$\kappa^{-1}d_{g_a}(x, z) \leq d_{g_b}(x, z) \leq \kappa d_{g_a}(x, z) \quad (33)$$

on held-out secants. Distortion multiplies along a chain of regions; large cycle distortion is evidence that no single global geometry is justified. The method then maintains an atlas of local quotients rather than averaging incompatible weights.

9 Systems design for ENNX

The ENNX `PackedSearch` backend realizes the quantized domain (2). Each logical variable block is stored as a contiguous sequence of 4- or 8-bit quantization codes together with a decoder scale, distance weight, and proposal radius; the current API names this record a `Leaf`. ENN distances are accumulated from decoded numerical differences and are therefore distinct from both bitwise Hamming distance and the code-coordinate distance d_H in (3).

Candidate generation is deterministic conditional on a centre, random seed, block metadata, and proposal length. The tuple of these values is an implicit candidate descriptor. In the lazy path, the selected descriptor is materialized as a full bit-packed vector only after acquisition. This reduces candidate-write traffic but does not alter the logical search space, proposal distribution, or statistical model.

Computational viability requires geometry uncertainty not to erase the throughput advantage of bit-packed low-precision data and accelerator-resident scoring. The current ENNX architecture provides the required outer loop: seed-defined candidate descriptors, resident bit-packed observations, decoded distance/top- k kernels, ENN prediction, UCB or Thompson acquisition, and trust-region state. The new cost is evaluating a small family of blockwise distance-weight vectors rather than one vector.

Let $J = |Q_t|$, $M = |A_t|$, N be resident history size, k the retained neighbour count, and s the coordinate-change budget. The complexity depends on the proposal operator. For a support-restricted proposal, after the base-to-observation distances have been cached for each geometry, each candidate distance is a rank- s coordinate update. A direct exact scorer then costs

$$O(JMN(s + k)). \quad (34)$$

For the regular full-vector proposal, distance scoring examines all D decoded coordinates and the corresponding bound is $O(JMN(D + k))$. Lazy materialization avoids writing all D output codes for unselected candidates, but does not remove the D work needed to score a full-vector proposal. Therefore a dimension-independent systems claim is valid only for the support-restricted path (or for a future index with a proved exact-recall shortcut), not for **PackedSearch** in general.

The representative count J is controlled by three mechanisms:

1. share seeds, selected coordinate sets, and decoded coordinate changes across geometries;
2. retain only prediction-distinct extreme geometries, not a dense parameter grid;
3. use column generation: score the current Q_t , then solve for a new geometry only at the few candidates whose robust acquisition can still improve.

An indexed shortlist replaces N by shortlist size, provided the shortlist has a reference-geometry recall guarantee. Candidate-value quantization and any future low-precision representation of geometry weights introduce different numerical errors and must be accounted for separately.

9.1 Incremental implementation

We separate the proposed implementation into independently testable components.

1. **Prediction surface.** The **PackedSearch** ask/tell interface remains unchanged, while the selected candidate’s ENN mean and error components are exposed for calibration.
2. **Finite geometry ensemble.** The ensemble interface accepts J blockwise distance-weight and proposal-radius vectors, generates candidates with shared seeds, scores their Cartesian product, and returns the maximizing configuration, geometry index, mean, uncertainty, and Ω_t .
3. **Reference coverage.** A configurable fraction of the candidate budget uses a fixed strictly positive reference geometry. UCB still selects the single objective query.
4. **Geometry updater.** The CPU control plane maintains the domination polytope and finite-model residual tests. LP solves occur once per BO round or region update, outside the distance kernel.
5. **Calibration.** Preassigned folds, replay evaluation, and interval-coverage telemetry support calibration; exact zero weights are disabled in the absence of a certificate.
6. **Continuous-set optimization.** Coordinate extremizers and acquisition-driven column generation extend the finite ensemble after its coverage and throughput have been established.

The existing all-data posterior-like draw remains an empirical Thompson option. The QR-ENN-UCB guarantee applies only after its draws or intervals are calibrated under adaptive data.

10 Related work

Unknown or implicit metric bandits already exist. Slivkins studies a taxonomy whose tree is known but whose numerical subtree widths are hidden [11]. Wanigasekara and Yu learn nonparametric similarities

among a finite set of contextual-bandit arms and adaptively partition the context–arm space [12]. Input warping learns bijective transformations for nonstationary GP BO [13], and BO with unknown stationary-kernel hyperparameters can retain no-regret guarantees by expanding the considered function class [14]. Misspecified GP bandits explicitly pay for uniform approximation error [15]. These results establish that learning an unknown metric is not itself new. QR-ENN-UCB differs by combining a pseudometric with nontrivial null fibres, confidence sets over local quotient geometries, an explicit geometry-disagreement term in acquisition and regret, and an ENNX-oriented systems design for quantized high-dimensional configurations. The same geometry controls candidate generation, neighbour retrieval, and trust-region shape, while the fibre approximation error remains explicit.

Related high-dimensional BO methods make different assumptions. REMBO uses a low-dimensional embedding [4]; BAXUS expands nested subspaces [6]; TuRBO localizes BO [3]; and cylindrical Thompson sampling changes proposal geometry [5]. QR-ENN-UCB instead tries to learn local equivalence classes from objective evidence.

11 Experimental protocol

The experimental protocol tests geometry identification, statistical calibration, optimization performance, and systems throughput separately before evaluating the complete method.

11.1 Synthetic benchmark families

The synthetic family fixes quotient dimension q while increasing ambient nuisance dimension $D - q$ without changing the quotient objective. It contains five settings:

1. axis-aligned exact null blocks, $\eta = 0$;
2. weakly active blocks with a controlled η floor;
3. correlated blocks for which (7) is nonunique;
4. quotient changes across regions, requiring an atlas;
5. rotated active subspaces inside a block, which the scalar block family cannot represent.

The fifth case is a negative control. Scalar block weights eliminate whole blocks but cannot recover an arbitrary low-rank subspace within a block. Representing that setting requires the extension $m_G(x, z) = (q(x) - q(z))^T G(q(x) - q(z))$ with block-low-rank G , together with its additional identification and systems costs.

The reported quantities are simple regret versus oracle calls and wall-clock time, coverage of η and ENN intervals, geometry-set coverage of $g_\star$, Ω_t , effective neighbour count, pool gap against a large offline reference pool, and candidate throughput. Results are stratified by D , q , J , noise level, and quantization format.

11.2 Ablations

At matched oracle-query and wall-clock budgets, we compare:

1. current single-geometry ENNX;
2. QR-ENN-UCB with the same point estimate but no geometry extremes;
3. robust prediction with Euclidean/reference candidate motion;
4. learned candidate motion with reference ENN distances;
5. full shared geometry;
6. TuRBO, BAXUS, cylindrical or spherical sampling, random search, and an appropriate direct-search baseline.

The two mismatch ablations isolate whether gains arise from better candidate coverage, better ENN neighbourhoods, or their coupling.

11.3 Evaluation criteria

Evidence against the proposed scaling mechanism consists of any persistent observation below:

1. empirical ENN intervals undercover materially under adaptive search;
2. $\mathcal{G}_t$ loses the planted geometry more often than its declared failure probability;
3. Ω_t remains linear in the regret accounting because geometries cannot be identified;
4. nuisance dimensions force J , pool size, or reference exploration to scale with D ;
5. the quotient metric improves prediction but harms candidate coverage, or conversely;
6. ensemble scoring removes the measured speed advantage over a GP or random-search baseline;
7. exact zero-weight selection improves benchmarks only because the benchmark’s null directions are known in advance.

12 Limitations and open problems

Proposition 8 is a correct finite-pool decision statement, but it is not yet a global no-regret theorem. A complete theory needs the following pieces.

Adaptive ENN concentration. A time-uniform result requires either confidence sequences for predictable local weights or a fold construction with exact independence. Nearest-neighbour sets selected after observing an adaptive trajectory are not automatically covered by a fixed-query concentration bound.

Geometry contraction. Sublinear cumulative regret requires assumptions under which $\Omega_t(x_t)$ is summable or sublinear; secant domination alone is insufficient. A promising route is a finite nested family of quotient models with time-uniform misspecification tests followed by regret-bound balancing [16, 17]. A continuous block metric may follow by covering the parameter set, but neighbour-order discontinuities must be handled.

Candidate coverage under adaptive geometry. Extending (32) to support-restricted proposals on quantized variables requires a coverage result for candidates generated under a mixture of learned metrics. The relevant dimension is likely a near-optimality or zooming dimension [10], not merely the number of active coordinates or nonzero block weights.

Misspecification adaptation. When η is unknown, a grid of misspecification budgets must balance invalid optimism at small η against vacuous bounds at large η . Phased elimination or regret balancing, analogous to misspecified kernel bandits [15], is a plausible mechanism.

Local-to-global accounting. A global result must account for trust-region restarts and metric transitions. A region graph with controlled overlap distortion may support an atlas theorem; arbitrary changing quotients will not.

Computational geometry. The number of extreme metrics required to approximate $\max_{g \in \mathcal{G}_t} u_{t,g}(x)$ after top- k neighbour identities are considered is not yet characterized. The score is piecewise smooth over cells of metric space induced by pairwise distance-order hyperplanes. An acquisition-driven traversal of only occupied cells may be much cheaper than a uniform metric net.

13 Relation to zeroth-order gradient estimation

The oracle in (1) is zeroth order, but QR-ENN-UCB is not a zeroth-order gradient algorithm. If a separate method draws u on a q -sphere and evaluates an antithetic pair at radius μ , the classical two-point estimator scales with q rather than D only when the quotient is valid. Under a residual $|e| \leq \eta$, its gradient bias contains a term of order $q\eta/\mu$. That creates a different optimization tradeoff and consumes two evaluations per direction [20, 21]. Such estimators may provide diagnostics, but they are outside the BO algorithm and regret analysis developed here.

14 Conclusion

We introduced a pure BO framework over two coupled unknowns: the maximizing configuration and the local geometry through which objective values can be predicted. ENN supplies the nonparametric surrogate, the quotient hypothesis provides a route to intrinsic-dimension dependence, and the geometry confidence set prevents unsupported elimination of directions. The resulting regret decomposition is

$$\begin{aligned} \text{regret} \lesssim & \text{pool coverage} + \text{ENN uncertainty in dimension } q \\ & + \text{geometry disagreement} + \text{fibre misspecification} + \text{region transitions.} \end{aligned}$$

The method scales only when these terms are simultaneously estimable and the geometry-disagreement term contracts without ambient-dimensional probing. The proposed ENNX design makes this condition empirically testable while retaining low-precision candidate-scoring throughput.

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